

THE EXACT DISTRIBUTION OF WILKS' CRITERION

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SUMMARY

By using the method of convolutions, Schatzoff (1966), Pillai and Gupta (1969) and Gupta (1971) obtained representations for the distribution of Wilks' criterion in MANOVA. In this article, we have shown that their results can be more easily obtained by the method of inverse Mellin transforms.

1. Introduction

Suppose that $X_{p \times n}$ ($n > p$) and $Y_{p \times q}$ are a pair of random matrices whose elements have joint probability density function (p.d.f.)

(1.1) $(2\pi)^{-p(n+q)/2} |\Sigma|^{-(n+q)/2} \exp \left[-\frac{1}{2} \text{tr} \Sigma^{-1} \{XX' + (Y-\mu)(Y-\mu)'\} \right]$,
where $\Sigma_{p \times p}$ is positive definite. The likelihood ratio statistic for testing the hypothesis $H: \mu = 0$, given by Wilks (1932) is

(1.2) $\Lambda_{p,q,n} = \det(A)/\det(A+B)$,

where $A = XX'$ and $B = YY'$. The above model arises in the multivariate analysis of variance, where p is the number of variates, A and B are the sums of squares and products matrices for error and hypothesis respectively, and n and q are the corresponding degrees of freedom.

Several authors have obtained the exact distribution of $\Lambda_{p,q,n}$ by using different techniques, see Mathai (1973). Since then a few relevant developments have taken place. These include methods for actual computation of exact percentiles and powers using convolutions (Hart and Money (1976)) and using the methods of mixtures (Tretter and Walster (1975)). These are listed in a very complete review by Pillai (1976, 1977).

By using the method of convolutions, Schatzoff (1966) has shown that under the null hypothesis, both the density and distribution functions of Λ have exact closed form representations when p or q is even. Schatzoff was able to obtain recurrence relations

among the coefficients which enabled him to compute percentage points and extend tables given earlier by Pillai (1960). Then, Pillai and Gupta (1969) used Schatzoff's representation of $(1 - \ln \Lambda)$ and obtained the exact densities in multiple sums for $p = 3, 4, 5, 6$. Also, they computed the percentage points and extended the tables of Schatzoff (1966). Gupta (1971) also used Schatzoff's representation of $(-\ln \Lambda)$ to obtain the exact distribution of Λ in the noncentral linear case, i.e. when the alternative hypotheses is of unit rank. He gave explicit expressions for the densities for $p = 2, 3, 4, 5$.

The purpose of this paper is to show that the results obtained by Schatzoff (1966), Pillai and Gupta (1969) and Gupta (1971) can be more easily obtained by the method of inverse Mellin transforms. In Sections 2 and 3, we have obtained their results in the central and noncentral linear cases respectively.

2. Exact distribution of $\Lambda_{p,q,n}$ in the null case

Anderson (1958) shows that the distribution of $\Lambda_{p,q,n}$ is the same as that of $\Lambda_{q,p,n+q-p}$. Hence, when obtaining the distribution of Λ , without loss of generality, we need to only consider the cases where $q \geq p$. It can be shown that the h -th moment of Λ under the null hypothesis is

$$(2.1) \quad E(\Lambda^h) = \prod_{j=1}^p \{ \Gamma[h + \frac{1}{2}(n+1-j)] \Gamma[\frac{1}{2}(n+q+1-j)] / \Gamma[h + \frac{1}{2}(n+q+1-j)] \Gamma[\frac{1}{2}(n+1-j)] \}.$$

However, for convenience, we shall consider the cases:

Case 1: p -even.

Let $p = 2m$. By applying the duplication formula for Gamma functions

$$(2) \quad 2^{2z-1} \Gamma(z) \Gamma(z + \frac{1}{2}) = \pi^{\frac{1}{2}} \Gamma(2z),$$

the h -th moment of Λ is

$$(2.3) \quad E(\Lambda^h) = \prod_{j=1}^m \{ \Gamma(2h+n+1-2j) \Gamma(n+q+1-2j) / \Gamma(2h+n+q+1-2j) \Gamma(n+1-2j) \}.$$

Applying, to the Gamma ratios in (2.3) the well-known (see e.g. Erdélyi (1953), p. 8) result

$$(2.4) \quad \Gamma(x) / \Gamma(x+y) = \{ \Gamma(y) \}^{-1} \sum_{j=0}^{\infty} (-1)^j \binom{y-1}{j} (x+j)^{-1}; \quad x > 0, y > 0,$$

where the series (2.4) reduces to a finite series whenever y is a positive integer, then (2.3) can be written as

$$(2.5) \quad E(\Lambda^h) = C \prod_{j_1=0}^{q-1} \dots \prod_{j_m=0}^{q-1} a(j_1, j_2, \dots, j_m) (2h+n+j_1-1)^{-1} (2h+n+j_2-3)^{-1} \dots (2h+n+j_m-2m+1)^{-1}$$

where

$$(2.6) \quad C^{-1} = \prod_{j=1}^m B(n+1-2j, q),$$

$$a(j_1, j_2, \dots, j_m) = \prod_{k=1}^m (-1)^{j_k} \binom{q-1}{j_k}.$$

By applying the method of inverse Mellin transforms, the p.d.f. of Λ is

$$(2.7) \quad f(\Lambda) = C \Lambda^{-1} \prod_{j_1=0}^{q-1} \dots \prod_{j_m=0}^{q-1} a(j_1, j_2, \dots, j_m) (2\pi i)^{-1} \int \Lambda^{-h} (2h+n+j_1-1)^{-1} \dots (2h+n+j_m-2m+1)^{-1} dh$$

where $i = (-1)^{\frac{1}{2}}$, and the contour integral in (2.7) is taken over some suitable contour. In order to evaluate the integral (2.7), we first have to determine the orders of the poles of the integrand. Since these are in turn determined by the values of $t_1=j_1-1$, $t_2=j_2-3$, ..., $t_m=j_m-2m+1$, we then have to split the multiple sum (2.7) by considering the constraints on the t_i :

$$(2.8) \quad t_1=t_2, t_1=t_3, \dots, t_1=t_m, t_1=t_2=t_3, \dots, t_1=t_2=\dots=t_m.$$

Since the form of the p.d.f. of Λ becomes complicated in the general case, we consider the special cases $p = 2, 4, 6$.

From (2.7), the p.d.f. of $\Lambda_{2,q,n}$ is

$$(2.9) \quad f(\Lambda) = [B(n-1, q)]^{-1} \Lambda^{-1} \prod_{j_1=0}^{q-1} a(j_1) (2\pi i)^{-1} \int \Lambda^{-h} (2h+n+j_1-1)^{-1} dh$$

$$= [2B(n-1, q)]^{-1} \Lambda^{(n-3)/2} \prod_{j_1=0}^{q-1} (-1)^{j_1} \binom{q-1}{j_1} \Lambda^{j_1/2}$$

$$= [2B(n-1, q)]^{-1} \Lambda^{(n-3)/2} (1-\Lambda)^{\frac{1}{2}} q^{-1},$$

which agrees with the result given by Anderson (1958).

When $p = 4$, (2.7) becomes

$$\begin{aligned}
 (2.10) \quad f(\Lambda) &= C\Lambda^{-1} \left\{ \sum_{j_1=0}^{q-3} a(j_1, j_1+2)(2\pi i)^{-1} \int \Lambda^{-h} (2h+n+j_1-1)^{-2} dh \right. \\
 &\quad \left. + \sum_{\substack{j_1=0 \\ j_1 \neq j_2-2}}^{q-1} \sum_{j_2=0}^{q-1} a(j_1, j_2)(2\pi i)^{-1} \int \Lambda^{-h} (2h+n+j_1-1)^{-1} (2h+n+j_2-3)^{-1} dh \right\} \\
 &= \left\{ \prod_{j=1}^2 2B(n+1-2j, q) \right\}^{-1} \Lambda^{(n-3)/2} \left\{ \sum_{j=0}^{q-3} \binom{q-1}{j} \binom{q-1}{j+2} (-\ell n \Lambda) \Lambda^{j/2} \right. \\
 &\quad \left. + 2 \sum_{\substack{j=0 \\ j \neq k-2}}^{q-1} \sum_{k=0}^{q-1} (-1)^{j+k} \binom{q-1}{j} \binom{q-1}{k} (\Lambda^{j/2} - \Lambda^{\frac{1}{2}k-1}) / (k-j-2) \right\},
 \end{aligned}$$

which agrees with the result given by Pillai and Gupta (1969).

In the case $P=6$, proceeding as before, the p.d.f. of $\Lambda_{6,q,n}$ results in the same expression as given by Pillai and Gupta (1969). For higher values of p , the exact p.d.f. can again be obtained using (2.7). However, the expressions become complicated.

Case 2: p -odd

Let $P = 2m+1$. By again applying the formulae (2.2) and (2.4) to (2.1), the h -th moment of Λ is

$$\begin{aligned}
 (2.11) \quad E(\Lambda^h) &= C_1 \sum_{j_1=0}^{q-1} \dots \sum_{j_m=0}^{q-1} \sum_{j_{m+1}=0}^{\infty} b(j_1, j_2, \dots, j_m, j_{m+1}) (2h+n+j_1-1)^{-1} \\
 &\quad \cdot (2h+n+j_2-3)^{-1} \dots (2h+n+j_m-2m+1)^{-1} (2h+n+2j_{m+1}-2m)^{-1}
 \end{aligned}$$

where

$$(2.12) \quad b(j_1, j_2, \dots, j_m, j_{m+1}) = \left\{ \prod_{k=1}^m (-1)^{j_k} \binom{q-1}{j_k} \right\} (-1)^{j_{m+1}} \binom{\frac{1}{2}q-1}{j_{m+1}}$$

$$C_1 = 2 \left\{ B\left(\frac{1}{2}(n+1-p), \frac{1}{2}q\right) \prod_{j=1}^m B(n+1-2j, q) \right\}^{-1},$$

and we note that the series in (2.11) indexed by j_{m+1} will reduce to a finite series with

upper limit of summation $\frac{1}{2}q-1$ whenever q is even.

By applying the method used in Case 1, the p.d.f. of Λ can be obtained. The results corresponding to $p = 3, 5$ are the same as obtained by Pillai and Gupta (1969).

3. Exact distribution of $\Lambda_{p,q,n}$ in the noncentral linear case

In the noncentral linear case, Kshirsagar (1961) has shown that $\Lambda_{p,q,n}$ is distributed as $\prod_{j=1}^p X_j$, where the X_j are mutually independent random variables, X_1 having p.d.f.

$$(3.1) [B(\frac{1}{2}n, \frac{1}{2}q)]^{-1} e^{-\frac{1}{2}\lambda^2} x_1^{\frac{1}{2}n-1} (1-x_1)^{\frac{1}{2}q-1} {}_1F_1(\frac{1}{2}(n+q); \frac{1}{2}q; \frac{1}{2}\lambda^2(1-x_1)),$$

$0 < x_1 < 1$; λ^2 being the single non-zero root of the determinantal equation

$$(3.2) |\mu\mu' - \delta^2\Sigma| = 0,$$

and X_j ($j=2, \dots, p$) has p.d.f.

$$(3.3) [B(\frac{1}{2}(n+1-j), \frac{1}{2}q)]^{-1} x_j^{(n-1-j)/2} (1-x_j)^{\frac{1}{2}q-1}, \quad 0 < x_j < 1.$$

Thus, the h -th moment of Λ is

$$(3.4) E(\Lambda^h) = K_1 \sum_{j=0}^{\infty} (\frac{1}{2}(n+q))^j \frac{(\frac{1}{2}\lambda^2)^j}{j!} \left\{ \frac{\Gamma(h+\frac{1}{2}n)}{\Gamma(h+\frac{1}{2}(n+q)+j)} \prod_{k=2}^p \frac{\Gamma[h+\frac{1}{2}(n+1-k)]}{\Gamma[h+\frac{1}{2}(n+q+1-k)]} \right\}$$

where

$$(3.5) K_1 = e^{-\lambda^2/2} \prod_{k=1}^p \{ \Gamma[\frac{1}{2}(n+q+1-j)] / \Gamma[\frac{1}{2}(n+1-j)] \}.$$

For convenience, we again consider the two cases:

Case 1: p -odd

Let $p = 2m+1$. Using (2.4) after applying (2.2) to the Gamma ratios in (3.4), the h -th moment of Λ is

$$(3.6) E(\Lambda^h) = K_2 \sum_{j=0}^{\infty} d_j \sum_{k_1=0}^{q-1} \dots \sum_{k_m=0}^{q-1} \sum_{k_{m+1}=0}^{\infty} c_j(k_1, k_2, \dots, k_{m+1}) (2h+n+k_1-2)^{-1} \\ \cdot (2h+n+k_2-4)^{-1} \dots (2h+n+k_m-2m)^{-1} (2h+n+2k_{m+1})^{-1}$$

where

$$c_j(k_1, k_2, \dots, k_m, k_{m+1}) = (-1)^{k_1 + \dots + k_{m+1}} \binom{q-1}{k_1} \dots \binom{q-1}{k_m} \binom{\frac{1}{2}q+j-1}{k_{m+1}},$$

$$(3.7) \quad d_j = \left(\frac{1}{2}(n+q)\right)_j \left(\frac{1}{2}\lambda^2\right)^j / \left(\frac{1}{2}q\right)_j!,$$

$$K_2 = 2e^{-\frac{1}{2}\lambda^2} \left\{ B\left(\frac{1}{2}n, \frac{1}{2}q\right) \prod_{j=1}^m B(n-2j, q) \right\}^{-1}.$$

Therefore, the p.d.f. of Λ can be obtained by using the method applied in Section 2. We have verified that the results corresponding to $P=3,5$ are the same as obtained by Gupta (1971).

Case 2: p -even

Let $p = 2m+2$. Proceeding as in the previous case, the h -th moment of Λ can be written in the form

$$(3.8) \quad E(\Lambda^h) = C_3 \sum_{j=0}^{\infty} d_j \sum_{k_1=0}^{q-1} \dots \sum_{k_m=0}^{q-1} \sum_{k_{m+1}=0}^{\infty} \sum_{k_{m+2}=0}^{\infty} c_j^* \\ (k_1, k_2, \dots, k_m, k_{m+1}, k_{m+2}) \\ \cdot (2h+n+k_1-2)^{-1} \dots (2h+n+k_m-2m)^{-1} (2h+n+2k_{m+1}-2m-1)^{-1} \\ \cdot (2h+n+2k_{m+2})^{-1}$$

where

$$(3.9) \quad C_3 = 4 \left\{ B\left(\frac{1}{2}n, \frac{1}{2}q\right) B\left(\frac{1}{2}(n+1-p), \frac{1}{2}q\right) \prod_{j=1}^m B(n-2j, q) \right\}^{-1},$$

$$c_j^*(k_1, k_2, \dots, k_m, k_{m+1}, k_{m+2}) = \left\{ \prod_{\ell=1}^m (-1)^{k_{\ell}} \binom{q-1}{k_{\ell}} \right\} (-1)^{k_{m+1}+k_{m+2}} \binom{\frac{1}{2}q-1}{k_{m+1}} \\ \binom{\frac{1}{2}q+j-1}{k_{m+2}}$$

Thus, the density function can be obtained as in the previous cases. For $p=2$ and $p=4$,

the results are the same as given in Gupta (1971).

4. Concluding remarks

It should be noticed that the method of inverse Mellin transforms given in this article has an advantage over the method of convolutions in that whereas the method of convolutions requires prior knowledge of the density function of $\Lambda_{p,q,n}$ for lower values of p , the method of inverse Mellin transforms obtains the density function directly for arbitrary p without any such knowledge. Finally, we note that the method presented here can be applied to various other multivariate test criteria. For details, we refer to Richards (1978).

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